3.8.6 Magnetic dipole moment of revolving electron

Suppose an electron undergoes circular motion around the nucleus as shown in Figure 3.36. The circulating electron in a loop is like current in a circular loop (since flow of charge constitutes current). The magnetic dipole moment due to current carrying circular loop is

$$\vec{\mu}_L = I\vec{A} \qquad (3.45)$$

In magnitude,

$$\mu_L = IA$$

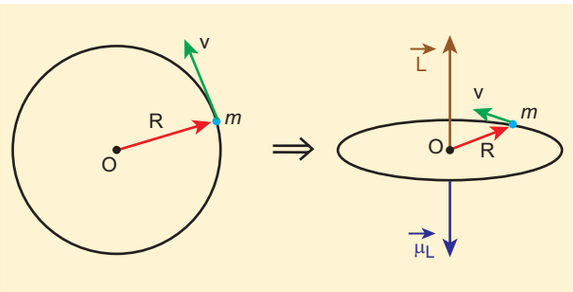


Figure 3.36 (a) Electron revolving in a circular orbit (b) Direction of magnetic dipole moment vector and orbital angular momentum vector are opposite

If T is the time period of revolution of an electron, the current due to circular motion of the electron is

$$I = \frac{-e}{T} \qquad (3.46)$$

where $-e$ is the charge of an electron. If R is the radius of the circular orbit and υ is the velocity of the electron in the circular orbit, then

$$T = \frac{2\pi R}{\upsilon} \qquad (3.47)$$

Using equation (3.46) and equation (3.47) in equation (3.45), we get

$$\mu_L = -\frac{e}{\frac{2\pi R}{\upsilon}}\pi R^2 = -\frac{e\upsilon R}{2} \qquad (3.48)$$

where $A = \pi R^2$ is the area of the circular loop. By definition, angular momentum of the electron about O is

$$\vec{L} = \vec{R} \times \vec{p}$$

In magnitude,

$$L = Rp = m\upsilon R \qquad (3.49)$$

Using equation (3.48) and equation (3.49), we get

$$\frac{\mu_L}{L} = -\frac{e\upsilon R/2}{m\upsilon R} = -\frac{e}{2m} \Rightarrow \vec{\mu}_L = -\frac{e}{2m}\vec{L} \qquad (3.50)$$

The negative sign indicates that the magnetic moment and angular momentum are in opposite direction.

In magnitude,

$$\frac{\mu_L}{L} = \frac{e}{2m} = \frac{1.60\times10^{-19}}{2\times9.11\times10^{-31}} = 0.0878\times10^{12}\ \text{C kg}^{-1}$$

$$\frac{\mu_L}{L} = 8.78\times10^{10}\ \text{C kg}^{-1} = \text{constant}$$

The ratio $\frac{\mu_L}{L}$ is a constant known as gyro-magnetic ratio $\left(\frac{e}{2m}\right)$. It must be noted that the gyro-magnetic ratio is a constant of proportionality which connects angular momentum of the electron and the magnetic moment of the electron.

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